TEXORA · LECTURE SHEET

Textile Finishing: Definition and Classification

The last stage of wet processing — where a dyed, printed cloth that is still harsh, creased and unstable is turned into a saleable fabric with the handle, appearance and performance the buyer specified.

WP-4201 · TEXTILE FINISHING · TOPIC 1 · SPECIALIZATION — WET PROCESSING

1. Definition of textile finishing

Textile finishing is the series of treatments applied to a fabric after dyeing or printing — and in a few cases to undyed fabric — to improve its appearance, handle (feel), dimensional stability and functional performance, and to make it fit for its intended end use. Finishing is sometimes defined more broadly to include every process after weaving or knitting, but in wet-processing practice the narrow, useful definition is this: *finishing begins where coloration ends.* Pretreatment (desizing, scouring, bleaching, mercerising) prepares the fabric for colour; dyeing and printing add colour; finishing gives the cloth its final character.

Two points in the definition matter. First, finishing is **end-use driven**: the same grey-dyed cotton can be finished as a soft, sanforized shirting, a stiff interlining, a water-repellent tent cloth or a flame-retardant furnishing — the finish, not the base fabric, decides the market. Second, finishing is **value-adding and largely irreversible in effect on value**: a poorly finished lot cannot be re-dyed into a good one, and most finishing faults (uneven softening, resin spots, residual shrinkage) are found only by the garment maker or the final consumer, when correction is impossible.

> **WORKING DEFINITION**
>
> Finishing = all mechanical and chemical treatments after coloration that (a) stabilise dimensions, (b) set the surface and handle, and (c) add aesthetic or functional properties — applied so that the fabric meets the buyer's specification for appearance, feel, care behaviour and service performance.

2. Objectives of finishing

- **Dimensional stabilisation:** control width, length, skew and bow; reduce residual shrinkage (for example to ≤1% in sanforized cotton) so garments keep their size after washing.
- **Appearance:** a flat, smooth, crease-free surface; controlled lustre (bright, matt or Schreiner lines); clear print definition; uniform width and shade across the piece.
- **Handle:** softness, fullness, stiffness, drape or crispness — the tactile quality the customer judges first, adjusted with softeners, stiffeners and mechanical action.
- **Easy care:** crease recovery, wash-and-wear behaviour, minimum ironing, and resistance to shrinkage and pilling in use.
- **Functional performance:** water repellency, flame retardancy, soil release, antistatic behaviour, antimicrobial protection, UV protection and moisture management.
- **Protection in storage and transit:** mothproofing, mildew resistance and rot-proofing for fabrics that will be warehoused or used outdoors.
- **Economy and saleability:** to present a uniform, defect-minimised product that passes inspection (shade, width, GSM, shrinkage, fastness) and commands its price.

3. Position in the processing sequence

The finishing department receives fabric from dyeing/printing and returns it to inspection, making-up and packing. Its position is fixed, because every finishing operation assumes that colour is complete and that no further hot wet treatment — which would strip a softener, relax a resin or re-shrink the cloth — will follow.

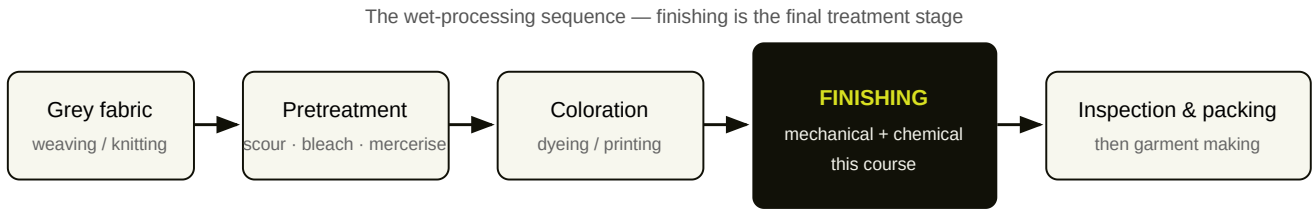


Figure 1 — Position of finishing in the processing sequence. No hot wet process follows finishing.

Within finishing itself the order is also logical: corrections and wet chemical application come first (washing-off, padding of agents), then drying, mechanical surface work (raising, shearing, calendering) as specified, then curing or heat setting where a reaction or a thermal set is needed, and finally conditioning, inspection and rolling. Topic 4 shows how the machines line up in that order.

4. Classification — an overview

Finishes are classified on several independent axes at once. A single production finish — resin easy-care finishing, for example — is simultaneously chemical, wet, durable and functional. Learning the axes separately, then combining them, is the correct way to describe any finish in an examination or a mill specification.

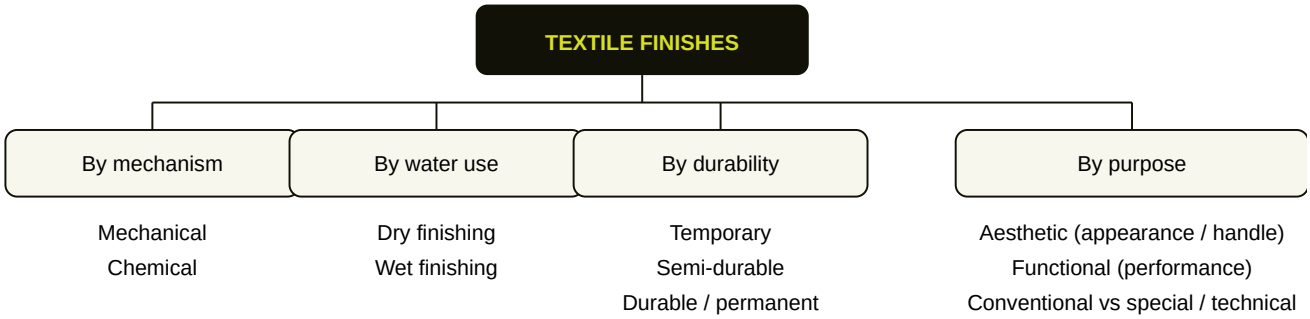


Figure 2 — Classification tree of textile finishes by mechanism, water use, durability and purpose.

5. Mechanical versus chemical finishing

ASPECT	MECHANICAL (PHYSICAL) FINISHING	CHEMICAL FINISHING
Agent of change	Heat, moisture, pressure, tension, friction and cutting action of machines	Chemical substances applied to the fibre/fabric, often reacted or cured in place
Examples	Calendering, sanforizing, raising, shearing, decatising, emerising, heat setting, compacting, stentering	Resin easy-care, softening, water-repellent, flame-retardant, soil-release, antimicrobial, antistatic finishes
What changes	Fabric geometry, surface structure, yarn/fabric arrangement, dimensions	Fibre surface chemistry, crosslinking inside the fibre, deposited films and coatings
Typical durability	Often partly lost in washing (calender lustre) unless the change is structural (sanforizing, heat setting are durable)	Ranges from one wash (starch) to the life of the garment (covalent resin crosslinks, fluorocarbon repellents)
Main controls	Machine settings: temperature, pressure, speed, overfeed, moisture	Recipe: concentration, pH, catalyst, pick-up, drying and curing conditions

Many production finishes are **combined**: a resin finish is applied chemically but needs stentering and curing machines; a calender effect on a resin-treated fabric lasts far longer than on untreated cloth. Topics 2 and 3 treat the two families separately, as the syllabus does, but the mill always runs them as one line.

6. Dry versus wet finishing; temporary versus durable

CLASS	MEANING	EXAMPLES
Dry finishing	Operations on dry (or only conditioned) fabric; no liquor application	Calendering, raising, shearing, decatising, emerising, inspection
Wet finishing	Operations involving water or liquor — padding, washing-off, wet decatising stages	Pad application of resin/softener, washing after curing, crabbing of wool
Temporary finish	Removed by the first wash or even by storage/steaming	Starch and CMC stiffening, calender lustre on untreated cotton, simple weighting
Semi-durable finish	Survives several washes, then fades gradually	Paraffin water repellents, non-reactive softeners, Pyrovatex-type treatments in early form
Durable / permanent finish	Withstands repeated laundering for the useful life of the article	DMDHEU crosslinking, THPC flame retardant, sanforizing, heat setting of polyester, fluorocarbon repellency

Durability is a **tested claim, not an intention**: a finish is called durable only after the specified property is re-measured after an agreed number of standard washes (commonly 20–50 home launderings for durable-press and flame-retardant workwear claims).

7. Aesthetic versus functional finishes

AESTHETIC FINISHES (APPEARANCE AND HANDLE)	FUNCTIONAL FINISHES (PERFORMANCE IN USE)
Calendering and Schreiner lustre; embossing; moiré	Water repellency and waterproof coating
Raising, emerising (peach) and sueding for surface feel	Flame retardancy
Softening, stiffening and fulling for handle and drape	Soil release and anti-soiling
Anti-crease appearance in wear (aesthetic side of resin finishing)	Antistatic, antimicrobial and anti-odour finishes
Delustring, optical brightening for whiteness	UV protection, moisture management, insect/moth proofing

The same agent can sit in both columns: silicone softener is aesthetic (handle) yet also improves sewability and tear strength — a functional gain. Examination answers should state the *primary*

purpose first and note secondary effects.

8. Conventional versus special (technical) finishes

Conventional finishes are the routine set applied to most apparel and household fabrics: stentering to width, softening, calendering, sanforizing, ordinary crease-resist treatment. They are high-volume, low-cost per metre and expected by every buyer. **Special or technical finishes** add a certified performance for a defined end use and are specified, tested and priced separately: flame-retardant workwear and curtains, antimicrobial hospital textiles, fluorocarbon oil-and-water repellency for outdoor wear, soil-release for table linen and uniforms, antistatic clean-room and fuel-handling clothing, and high-visibility or phase-change technical textiles. The commercial distinction matters: a conventional finish is part of the base price; a special finish is a line item with a test standard attached.

9. Finishing routes for typical fabrics

STAGE	COTTON SHIRTING (DYED)	POLYESTER / POLYESTER-COTTON FABRIC
1	Wash-off after dyeing; hydroextract	Wash-off / reduction clear; hydroextract
2	Pad resin + catalyst + softener (Topic 3)	Pad softener / thermosol auxiliaries as required
3	Stenter dry and width-set	Heat set on stenter, typically 180–200 °C for polyester
4	Cure (resin crosslinking)	— (no resin cure unless easy-care blend)
5	Calender if lustre specified	Calender or emerise if specified
6	Sanforize (residual shrinkage ≤1%)	Final stenter width and overfeed control
7	Condition, inspect, roll	Condition, inspect, roll

Knitted fabrics follow a parallel route in which open-width or tubular compacting replaces sanforizing, and softening is often applied by exhaust in the dyeing machine before drying — the classification axes are unchanged, only the machinery differs (Topic 4).

10. Glossary — key terms

- **Finishing:** post-coloration mechanical and chemical treatments that give fabric its final appearance, handle, stability and performance.
- **Handle (hand):** the total tactile impression — softness, fullness, stiffness, smoothness and drape — judged by touch.

- **Residual shrinkage:** shrinkage still remaining in a finished fabric, measured by a standard wash test.
- **Durable finish:** a finish that retains its specified effect through repeated laundering for the article's useful life.
- **Conventional finish:** routine, expected treatment included in normal processing; **special finish:** separately specified and tested performance finish.
- **End use:** the final application of the fabric, which determines the finishing recipe and tests.

11. Key points

- Finishing starts where coloration ends; nothing hot and wet follows it.
- Classify any finish on four axes: mechanism, water use, durability, purpose — plus conventional versus special.
- Durability is proved by re-testing after standard washes, not claimed from the recipe.
- The finishing route is written for the fibre and the end use: sanforize cotton, heat-set polyester.

12. Summary

Finishing converts coloured cloth into a specified product. Its objectives — stability, appearance, handle, easy care and function — are delivered by mechanical and chemical means, in dry or wet form, with effects ranging from temporary starch to permanent crosslinking and heat setting. The classification framework of this sheet is the map for the rest of WP-4201: Topic 2 works through the mechanical family, Topic 3 the chemical family, and Topic 4 the machines and agents that deliver both.

13. Practice

1. Classify each of the following on all four axes (mechanism, water use, durability, purpose): (a) sanforizing, (b) starch stiffening, (c) fluorocarbon water repellent, (d) calendering of untreated cotton.
2. A buyer asks for a cotton fabric that "will not shrink in washing and will need little ironing". Name the two finishing operations you would specify and the class each belongs to.
3. Why must resin curing and heat setting come after dyeing, never before it?
4. Distinguish conventional from special finishes with two examples of each, and state how each is normally paid for.

Answer key

1. (a) Mechanical, wet-assisted (moisture + heat), durable, functional (stability). (b) Chemical, wet, temporary, aesthetic/handle (stiffness). (c) Chemical, wet (pad-cure), durable, functional. (d) Mechanical, dry, temporary, aesthetic (lustre).
2. Sanforizing (mechanical, durable — controls residual shrinkage, target ≤1%) and DMDHEU resin easy-care finishing (chemical, durable — gives crease recovery and wash-and-wear behaviour).
3. Curing/setting permanently fixes dimensions and fibre state. Dyeing afterwards involves hot liquor, swelling and mechanical action that would remove or distort the set, strip uncured deposits and create unlevel dyeing on crosslinked fibre; colour must therefore be complete and washed off first.
4. Conventional: stentering to width, softening — included in the base processing price. Special: flame-retardant finishing to a named standard, antimicrobial finishing with a certified test — quoted as separate line items because agents, testing and liability cost more.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib